Conditional Probability and Bayes Rule

Section 1: Joint Probability and Independence

1 If two events A and B are statistically independent with $P(A) = 0.4$ and $P(B) = 0.3$, find the joint probability $P(A \cap B)$.

a. 0.12

b. 0.70

c. 0.10

d. 0.30

2 Given $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cup B) = 0.7$, find $P(A \cap B)$.

a. 0.20

b. 0.10

c. 0.30

d. 0.15

3 Are the events in question 2 statistically independent?

a. Yes, because $P(A)P(B) = 0.5 \times 0.4 = 0.20 = P(A \cap B)$

b. No, because $P(A) \neq P(B)$

c. No, because $P(A \cup B) \neq 1$

d. Cannot be determined without conditional probabilities

Section 2: Conditional Probability and the Chain Rule

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4 Given $P(A) = 0.6$ and $P(A \cap B) = 0.18$, calculate the conditional probability $P(B|A)$.

a. 0.30

b. 0.108

c. 0.42

d. 0.25

5 In an autoregressive language model, write the joint probability of a sequence of three tokens $P(w_1, w_2, w_3)$ using the probability Chain Rule.

a. $P(w_1)P(w_2|w_1)P(w_3|w_1, w_2)$

b. $P(w_1) + P(w_2) + P(w_3)$

c. $P(w_1)P(w_2)P(w_3)$

d. $P(w_3|w_2)P(w_2|w_1)$

6 If $P(A|B) = 0.8$ and $P(B) = 0.25$, find $P(A \cap B)$.

a. 0.20

b. 0.55

c. 0.3125

d. 0.10

Section 3: Law of Total Probability

7 Factory line 1 produces 60% of microchips with defect rate 2%. Line 2 produces 40% with defect rate 5%. What is the total probability that a random chip is defective?

a. 0.032

b. 0.070

c. 0.035

d. 0.025

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8 In question 7, if a selected chip is defective, what is the probability it came from Line 2?

a. $\frac{0.020}{0.032} = 0.625$

b. 0.400

c. 0.375

d. 0.500

9 State the Law of Total Probability for an exhaustive partition $\{A_1, A_2, \ldots, A_k\}$ of the sample space.

a. $P(B) = \sum_{i=1}^{k} P(B|A_i)P(A_i)$

b. $P(B) = \prod_{i=1}^{k} P(B|A_i)$

c. $P(B) = \max_i P(B|A_i)$

d. $P(B) = \sum_{i=1}^{k} P(A_i|B)$

Section 4: Bayes Theorem in Machine Learning and Screening

10 In medical screening, a disease has prevalence $P(D) = 0.01$. The test has sensitivity $P(+|D) = 0.99$ and false positive rate $P(+|\neg D) = 0.02$. What is the posterior probability $P(D|+)$ that a positive patient has the disease?

a. $\approx 33.3\%$

b. 99.0%

c. 50.0%

d. 5.0%

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11 What is the foundational simplifying assumption of the Naive Bayes text classifier?

a. Features (words) are conditionally independent given the class label

b. All prior probabilities are equal

c. The vocabulary size is infinite

d. Features follow a uniform distribution

12 In Maximum A Posteriori (MAP) estimation, how does the optimization objective differ from standard Maximum Likelihood Estimation (MLE)?

a. MAP incorporates a prior distribution: $\theta_{MAP} = \arg\max[\ln P(D|\theta) + \ln P(\theta)]$

b. MAP ignores all observed training data

c. MAP only applies to linear models

d. MAP minimizes variance instead of error

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